

Pattern Matching and Machine Learning for Audio Signal Processing

Exercise sheet 4

To be uploaded in eCampus till: 17-05-2026 22:00 (strict deadline)

Exercise 4.1

[3 + 3 = 6 points]

Let $\text{DFT}_N = (\omega_N^{kj})_{j,k \in [0:N-1]}$ with $\omega_N := \exp(-2\pi i/N)$ be the DFT matrix of size N . For $x, y, z \in \mathbb{C}^N$ let $X := \text{DFT}_N \cdot x$, $Y := \text{DFT}_N \cdot y$, and $Z := \text{DFT}_N \cdot z$.

- (a) Show that if $x \in \mathbb{R}^N$, $N = 2M$, is a real signal, then X_0 and X_M are real, too, and $\forall k \in [1 : M - 1]$ it holds that $X_{N-k} = \overline{X_k}$, i.e. $\text{Re}(X_{N-k}) = \text{Re}(X_k)$ and $\text{Im}(X_{N-k}) = -\text{Im}(X_k)$.
- (b) For $x, y \in \mathbb{R}^N$, show how one can reconstruct the Fourier transforms X and Y from the Fourier transform Z of the complex signal $z := x + iy$. In other words: One can calculate the Fourier transform of two real signals using one complex Fourier transform.
(Hint: write $Z_k + Z_{N-k}$ and $Z_k - Z_{N-k}$ in terms of X and Y).

Exercise 4.2

[4 + 4 = 8 points]

In this exercise we discuss the omitted details of our audio denoising example. The scenario is as follows: a 4 Hz sine wave $\mathbf{a}$ is our source signal: $\mathbf{a} = (\sin(2\pi 4k/N))_{k \in [0:N-1]}$, where $N = 512$. This source signal is corrupted by (real-valued) additive noise, resulting in the noisy signal $\mathbf{b} = \mathbf{a} + \mathbf{r}$. Let $\mathbf{A}$, $\mathbf{R}$, and $\mathbf{B}$ denote the corresponding spectra, thus $\mathbf{A} = \text{DFT}_N \mathbf{a}$, etc. Typically, the magnitude spectrum $|\mathbf{R}|$ of such a noise signal is flat, whereas the energy of the source signal $\mathbf{a}$ is concentrated on a few frequencies. Suppose we have chosen a threshold $\theta > 0$ such that $|R(k)| \leq \theta$, for all $k \in [0 : N - 1]$ and $|A(k)| > 2\theta$, for all nonzero $A(k)$. By hard thresholding we set any coefficient in $\mathbf{B}$ whose absolute value is $\leq \theta$ to zero. All other values of $\mathbf{B}$ remain unchanged. The resulting spectral vector is denoted by $\mathbf{B}_\theta$.

- (a) Compute $\mathbf{A}$.
- (b) Compute both the denoised signal $\tilde{\mathbf{a}} := \text{DFT}_N^{-1} \mathbf{B}_\theta$ and the remaining difference signal $\tilde{\mathbf{a}} - \mathbf{a}$.